

From Disclosure to Topology: A Temporal Multiplex Framework for Extracting Firm Interdependence from 10-K Filings

[Author(s) removed for blind review]

Motivation. Firms are embedded in overlapping systems of rivalry, product substitution, and shared exposure, yet text-based firm networks in finance are typically single-layer similarity graphs (TNIC [5]) or firm-level scores [6]. SEC Form 10-K filings are an annually refreshed, mandated corpus in which firms self-report *distinct* relation types in distinct sections. We treat the 10-K as a generative substrate for a node-aligned *temporal multiplex* [1,2,4] and ask: **how do layer-specific topology and interlayer coupling co-evolve as the information environment shifts?** We contribute (i) a schema mapping 10-K Items onto node-aligned layers with financial node attributes, (ii) explicit edge-formation rules, including rules for undisclosed competitors and asynchronous fiscal calendars, and (iii) a hand-curated pilot on Alphabet (G), Meta (M), and Nvidia (N), FY2021–2025, that surfaces the pathologies an automated pipeline must resolve.

Positioning. Finance extracts relations from filings but collapses them into one layer (text similarity [5], rivalry-word counts [6], customer links [8]); network science offers multilayer formalism and diagnostics [1–4] but rarely confronts regulatory text, where extraction itself determines topology. We make that step auditable: every edge records its source Item, formation rule, and alignment convention.

Multiplex construction. Let $\mathcal{M}_t = \{G_t^{(\alpha)} = (V, E_t^{(\alpha)}, \mathbf{x}_t)\}_{\alpha}$ share node set V and node attributes $\mathbf{x}_{i,t}$ drawn from Item 8 segment notes: total revenue, segment revenue-share vector $\mathbf{s}_i$, long-lived-asset geography, and major-customer concentration. **L1 Product** (Items 1, 7): each firm maps to a set C_i over a 9-category taxonomy (advertising, devices, XR, cloud/data-center compute, AI platforms, content & streaming, enterprise software, autonomous mobility, health) with breadth weight $J_{ij} = |C_i \cap C_j|/|C_i \cup C_j|$; mapping segment revenues onto the same taxonomy gives a depth weight $R_{ij} = \sum_k \min(s_{ik}, s_{jk}) / \sum_k \max(s_{ik}, s_{jk})$. **L2 Competition** (Item 1): arc $i \rightarrow j$ if i names j ; a *proxy-injected* arc is added when i lists a competitor category that j ’s L1 set satisfies. **L3 Risk** (Item 1A): a bipartite firm–risk-theme graph (21 coded themes) projected with Jaccard weights. A sentiment layer (Item 7 embeddings) is deferred. Layers differ in directionality and edge semantics; we summarize $\mathcal{M}_t$ with the edge overlap $o_{ij} = \sum_{\alpha} a_{ij}^{(\alpha)}$, the fraction of dyads active in ≥ 2 layers, and Spearman correlation of node strength across layers [3]. Temporal change is measured per layer as $\Delta_t^{(\alpha)} = 1 - \text{sim}(E_t^{(\alpha)}, E_{t-1}^{(\alpha)})$, separating edge turnover from weight drift.

Nvidia’s attributes move most over 2021–2024 (period-aligned): revenue rises from \$26.9B to \$130.5B, its largest segment flips from Gaming (46%) to Data Center (88%), and the US share of its long-lived assets falls from 73% to 58%, while Alphabet and Meta stay above 87% advertising-led revenue. Using financials as attributes rather than a fifth layer avoids double-counting L1 and lets attributes condition edge formation.

Coding protocol. From 15 filings we recorded product, competitor, and risk outlines plus segment-note financials. A category is assigned only for a named revenue-generating offering; a risk theme only for a dedicated risk factor or explicit regulatory reference; a proxy arc $i \rightarrow j$ iff a category listed by i matches $C_{j,t}$ under the same mapping table.

Asynchronous clocks and structural breaks. Nvidia’s fiscal year ends in late January, so FY t covers roughly calendar $t-1$, lagging G and M by ≈ 11 months. Following the greatest-calendar-overlap convention, we *period-align* Nvidia FY $t+1$ with G/M FY t (common window 2021–2024) and treat filing-label alignment as a robustness check. Financial attributes also carry disclosure breaks that must be modeled, not ignored: Alphabet’s 2023 segment recast (DeepMind to Alphabet-level costs), Nvidia’s Data Center split into Compute and Networking, and ASU 2023-07 segment-expense reporting from FY2024.

Extraction pathologies. (a) *Naming collapse*: in 2021 Alphabet named 24 rivals and Meta 8 (each naming the other); from 2022 both disclose only categories, consistent with counsel advising generic competitive descriptions amid renewed antitrust scrutiny of interlocks. The explicit L2 graph falls from 3 arcs to 1 (N→G), while Nvidia names 18–19 rivals yearly. Without a documented proxy rule, L2 dynamics would reflect legal drafting, not competition. (b) *Semantic drift*: Alphabet adds “AI products and services” (2023), Meta “frontier AI models” (2025), Nvidia Arm-based CPU suppliers (FY2023). (c) *Cross-item redundancy*: Item 1 feeds both L1 and L2, requiring disjoint criteria (*what a firm sells* vs. *whom it claims to face*). (d) *Sticky text*: Item 1A is heavily carried forward [7], so onset dates bound, rather than measure, when a risk became salient.

Preliminary observations (Fig. 1; descriptive, $|V| = 3$). *L1 breadth vs. depth disagree*. Mean J rises from 0.27 to 0.39 as firms enter shared AI and cloud categories, but revenue-weighted R does not converge: G–M falls (0.86→0.77) as Cloud grows within Alphabet, G–N rises only from 0.04 to 0.07, and M–N is 0 throughout because AI is not a separately monetized line at Meta and sits inside “Data Center” at Nvidia. Convergence is thus a category-entry signal,

not a revenue-mix signal, and the two weights should be retained as distinct edge attributes. The M–N null overlap coexists with Nvidia’s rising customer concentration (three customers $\geq 10\%$ of revenue in FY2025), pointing to a vertical rather than horizontal tie [8]. *L2*: the proxy rule adds $G \rightarrow N$ from 2023, so the G – N dyad becomes reciprocal only after Alphabet’s AI competitor category appears. *L3*: a dedicated AI risk factor appears for Nvidia, then Alphabet (FY2023), then Meta (FY2024, citing the EU AI Act); an indebtedness risk leaves Nvidia’s filing by FY2025 and enters Alphabet’s in FY2025 after its bond issuance. *Beyond the cohort*: competitor mentions reach 41 external firms, 8 named by ≥ 2 cohort firms (Amazon and Microsoft by all three), marking the boundary for scaling V .

Why the disagreement matters. J and R are two projections of one firm–category bipartite graph: J tracks strategic entry, R economic mass. Entry should lead mass, so a widening J – R gap is a candidate early signal of structural change, and any result that flips between projections should be reported as projection-dependent. *Limitations*: with $|V| = 3$ we report no centrality or community results; taxonomy mapping is lossy (AI revenue is not separately reported); coding used one annotator.

Path to scale. An LLM/NER pipeline over S&P 500 10-Ks will use the pilot’s taxonomy, proxy, and alignment rules as annotated ground truth. With $|V| \approx 500$ we will test whether interlayer overlap and breadth–depth divergence grow over time and whether L3 onsets precede L1 entry under both alignments, against degree-preserving layer-shuffled nulls, and add a directed customer–supplier layer. Pre-specified hypotheses: **H1** changes in J lead changes in R ; **H2** risk-theme adoption precedes entry into the associated product category; **H3** the post-2021 decline in explicit L2 arcs is disclosure-driven, so proxy arcs recover a density-stable competition graph.

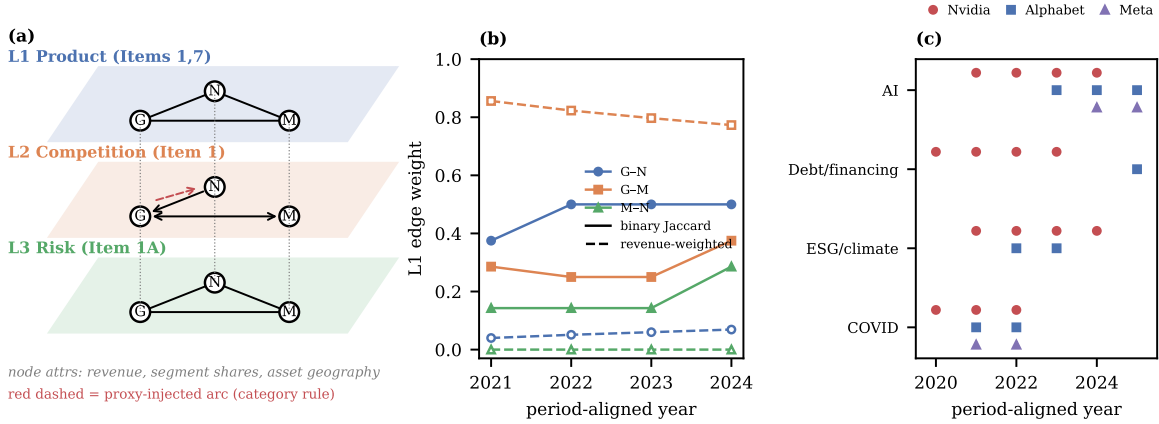


Figure 1: (a) Node-aligned multiplex for G, M, N: L1 product, L2 competition (black: named; red dashed: proxy-injected), L3 risk; Item 8 financials enter as node attributes. (b) L1 edge weights, period-aligned: binary category Jaccard (solid) vs. revenue-weighted Ruzicka (dashed). (c) Disclosure years of selected Item 1A risk themes.

References. [1] M. Kivelä et al., *J. Complex Netw.* 2(3):203–271, 2014. [2] S. Boccaletti et al., *Phys. Rep.* 544(1):1–122, 2014. [3] F. Battiston, V. Nicosia, V. Latora, *Phys. Rev. E* 89:032804, 2014. [4] M. De Domenico et al., *Phys. Rev. X* 3:041022, 2013. [5] G. Hoberg, G. Phillips, *J. Polit. Econ.* 124(5):1423–1465, 2016. [6] F. Li, R. Lundholm, M. Minnis, *J. Account. Res.* 51(2):399–436, 2013. [7] L. Cohen, C. Malloy, Q. Nguyen, *Lazy prices*, *J. Finance* 75(3):1371–1415, 2020. [8] L. Cohen, A. Frazzini, *Economic links and predictable returns*, *J. Finance* 63(4):1977–2011, 2008.